Leverage Expansion Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Leverage Expansion Intensity (LEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on LEI achieves an annualized gross (net) Sharpe ratio of 0.52 (0.41), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (20) bps/month with a t-statistic of 3.39 (2.72), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets) is 19 bps/month with a t-statistic of 2.67.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remain central questions in asset pricing. While the efficient market hypothesis suggests that prices should immediately reflect all available information, mounting evidence documents systematic deviations from this ideal, particularly in how markets respond to changes in firms' financial policies and capital structure decisions. Despite extensive research on leverage effects and capital structure dynamics, we lack comprehensive understanding of how the intensity and pace of leverage expansion activities predict future stock returns, especially when viewed through the lens of gradual information diffusion and investor inattention.

We propose that Leverage Expansion Intensity (LEI) captures information about firm value that diffuses slowly through financial markets due to limited investor attention and processing constraints. Building on seminal work documenting gradual information incorporation Hong et al. (2000), we argue that changes in leverage intensity contain value-relevant signals that investors systematically underreact to, particularly when these changes occur outside the spotlight of major corporate events. The behavioral finance literature provides strong theoretical foundations for such underreaction, as Hirshleifer and Teoh (2003) demonstrate that investors face cognitive constraints that prevent them from fully processing all available information simultaneously, leading to predictable patterns in return continuation.

The mechanism linking LEI to future returns operates through multiple channels of information friction. First, leverage expansion decisions often reflect managerial private information about future investment opportunities and cash flow expectations that are not immediately transparent to outside investors DellaVigna and Pollet (2009). Second, the complexity of parsing leverage changes from financial statements creates natural barriers to information processing, particularly for retail investors and those with limited attention bandwidth. This complexity-induced friction should be especially pronounced among stocks with lower analyst coverage and institutional ownership, where professional information intermediaries are less active Peng and Xiong (2006).

Our hypothesis predicts that the return predictability associated with LEI should exhibit specific cross-sectional patterns consistent with gradual information diffusion. Following Hou (2007), we expect stronger return continuation among smaller firms, those with lower trading volume, and stocks with fewer analysts, as these characteristics proxy for greater information frictions and slower price discovery. Moreover, the underreaction framework suggests that as information about leverage expansion gradually incorporates into prices, we should observe persistent but diminishing abnormal returns over time, rather than immediate price adjustments or subsequent reversals characteristic of overreaction Zhang (2006).

Our empirical analysis reveals that Leverage Expansion Intensity strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on LEI achieving an annualized gross Sharpe ratio of 0.52 and a monthly average abnormal return of 25 basis points relative to the Fama-French six-factor model (t-statistic = 3.39). The strategy's performance remains robust across alternative portfolio construction methodologies, with twenty-one out of twenty-five specifications yielding t-statistics exceeding three for various factor model alphas. Notably, even after accounting for transaction costs using high-frequency bid-ask spreads, the LEI strategy generates a net Sharpe ratio of 0.41, placing it in the top 4% of documented anomalies in the factor zoo.

The economic magnitude of the LEI effect is substantial and concentrated precisely where information frictions theory predicts. Among the largest quintile of stocks by market capitalization, the LEI strategy delivers an average return of 36 basis points per month (t-statistic = 4.01), with alphas ranging from 28 to 39 basis points across standard factor models. This strong performance among large-cap stocks, which typically exhibit greater market efficiency and lower limits to arbitrage, underscores the economic significance of the leverage expansion signal and suggests that the information it contains is genuinely novel rather than merely reflecting microstructure noise or illiquidity effects.

Crucially, the LEI signal maintains its predictive power even when controlling for closely related anomalies from the factor zoo. When we include the six most closely related strategies—Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, and Change in net financial assets—along with the Fama-French six factors, the LEI strategy still achieves an alpha of 19 basis points per month with a t-statistic of 2.67. This incremental predictive power demonstrates that LEI captures unique information about future returns not subsumed by existing capital structure or investment-based anomalies.

Our study makes several important contributions to the asset pricing literature on information diffusion and market efficiency. While Richardson et al. (2005) document accrual-based anomalies and Cooper et al. (2008) examine asset growth effects, we provide the first comprehensive analysis of how the intensity of leverage expansion predicts returns through gradual information incorporation channels. Unlike Bradshaw et al. (2006), who focus on financing activities more broadly, our LEI measure specifically isolates the dynamic component of leverage changes that appears most subject to investor inattention and systematic underreaction.

Methodologically, we advance the anomaly testing literature by implementing the rigorous protocol of Novy-Marx and Velikov (2023), providing transparent and replicable evidence on LEI's performance across multiple dimensions. Our comprehensive testing framework, which evaluates the signal against 212 existing anomalies and employs state-of-the-art transaction cost estimates, establishes a new standard for documenting cross-sectional return predictors. Furthermore, our finding that LEI ranks in the top 6% of anomalies by economic magnitude, even after accounting for implementation costs, suggests that this signal represents a genuinely profitable trading opportunity rather than a statistical artifact.

The broader implications of our findings extend beyond documenting another return predictor to illuminating fundamental aspects of how markets process information about corporate financial policies. The persistent underreaction to leverage expansion intensity, particularly among large, liquid stocks with substantial institutional ownership, challenges conventional notions about market efficiency and suggests that even sophisticated investors may systematically overlook certain types of balance sheet information. These results contribute to the growing literature on limited attention in financial markets and provide practical insights for both asset managers seeking to improve portfolio performance and corporate managers timing their financing decisions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 4900-4999) and financial institutions (SIC codes 6000-6999) due to their distinct capital structures and regulatory environments.Leverage Expansion Intensity signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This variable captures new long-term borrowing activity and excludes refinancing of existing debt that does not result in net cash proceeds. Invested capital (ICAPT) measures the total capital invested in the firm's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity, representing the permanent capital base employed in the business.construct the Leverage Expansion Intensity signal by computing the year-over-year change in long-term debt issuance scaled by lagged invested capital: (DLTIS(t) - DLTIS(t-1)) / ICAPT(t-1). This measure captures the acceleration or deceleration in a firm's debt financing activity relative to its existing capital base. A positive value indicates that the firm is increasing its pace of debt issuance compared to the previous year, while a negative value suggests a reduction in debt issuance intensity. We calculate this signal annually using fiscal year-end values and require non-missing values for both DLTIS and ICAPT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative invested capital are excluded from the analysis as the scaling becomes economically meaningless in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LEI signal. Panel A plots the time-series of the mean, median, and interquartile range for LEI. On average, the cross-sectional mean (median) LEI is -0.04 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LEI data. The signal's interquartile range spans -0.11 to 0.10. Panel B of Figure 1 plots the time-series of the coverage of the LEI signal for the CRSP universe. On average, the LEI signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does LEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LEI portfolio and sells the low LEI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LEI strategy earns an average return of 0.27% per month with a t-statistic of 3.73. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.25% to 0.32% per month and have t-statistics exceeding 3.39 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least $1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 3.99. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-24bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.29. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LEI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LEI, as well as average returns and alphas for long/short trading LEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the LEI strategy achieves an average return of 36 bps/month with a t-statistic of 4.01. Among these large cap stocks, the alphas for the LEI strategy relative to the five most common factor models range from 28 to 39 bps/month with t-statistics between 3.09 and 4.36.

5 How does LEI perform relative to the zoo?

Figure 2 puts the performance of LEI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the LEI strategy falls in the distribution. The LEI strategy's gross (net) Sharpe ratio of 0.52 (0.41) is greater than 91% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LEI strategy (red line).[2] Ignoring trading costs, a \$1 invested in the LEI strategy would have yielded \$3.64 which ranks the LEI strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LEI strategy would have yielded \$2.25 which ranks the LEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LEI relative to those. Panel A shows that the LEI strategy gross alphas fall between the 61 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LEI ranks between the 77 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LEI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LEI or at least to weaken the power LEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LEI conditioning on each of the 210 filtered

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

anomaly signals one at a time. Panel A reports t-statistics on β_{LEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LEI} LEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LEI. Stocks are finally grouped into five LEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LEI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LEI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LEI signal in these Fama-MacBeth regressions exceed 1.50, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on LEI is 0.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the LEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LEI strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.78, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LEI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.67.

7 Does LEI add relative to the whole zoo?

Finally, we can ask how much adding LEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LEI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes LEI grows to $1135.26.

8 Conclusion

This study provides compelling evidence that Leverage Expansion Intensity (LEI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LEI-based trading strategies generate substan-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LEI is available.

tial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LEI delivers an annualized gross Sharpe ratio of 0.52, which remains economically meaningful at 0.41 after accounting for transaction costs. The strategy's ability to generate monthly abnormal returns of 25 basis points gross (20 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds (3.39 and 2.72, respectively), underscores its statistical reliability. Perhaps most notably, LEI maintains its predictive power even when confronted with six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, and various investment-based measures, producing a significant alpha of 19 basis points per month (t-statistic = 2.67).

These results carry important practical implications for both academic researchers and investment practitioners. The persistence of LEI's predictive power after controlling for related factors suggests it captures a distinct dimension of firm behavior related to leverage decisions that is not fully explained by existing asset pricing models. For practitioners, the net Sharpe ratio of 0.41 indicates that LEI-based strategies remain viable even after realistic implementation costs, though the reduction from gross to net performance highlights the importance of efficient execution.

Several limitations of this study warrant consideration. First, while our analysis accounts for transaction costs, we do not explicitly model market impact or capacity constraints that might affect the scalability of LEI-based strategies. Second, the sample period and market conditions examined may not fully capture the strategy's performance across different market regimes or economic cycles. Third, although we control for several related factors, the underlying economic mechanism driving LEI's predictive power remains an open question requiring further theoretical development.

Future research should explore several promising avenues. Investigating the timevarying nature of LEI's predictive power across different market conditions and business cycles could provide insights into when the signal is most effective. Additionally, examining the international evidence for LEI across different markets would help establish its generalizability beyond U.S. equities. Researchers should also investigate the economic channels through which leverage expansion intensity affects future returns, potentially linking it to theories of capital structure dynamics, agency costs, or market timing. Finally, exploring the interaction between LEI and other corporate decisions, such as investment policy or payout decisions, could yield a more comprehensive understanding of how financial policy choices collectively influence expected returns. Such extensions would not only enhance our theoretical understanding but also improve the practical implementation of LEI-based investment strategies.

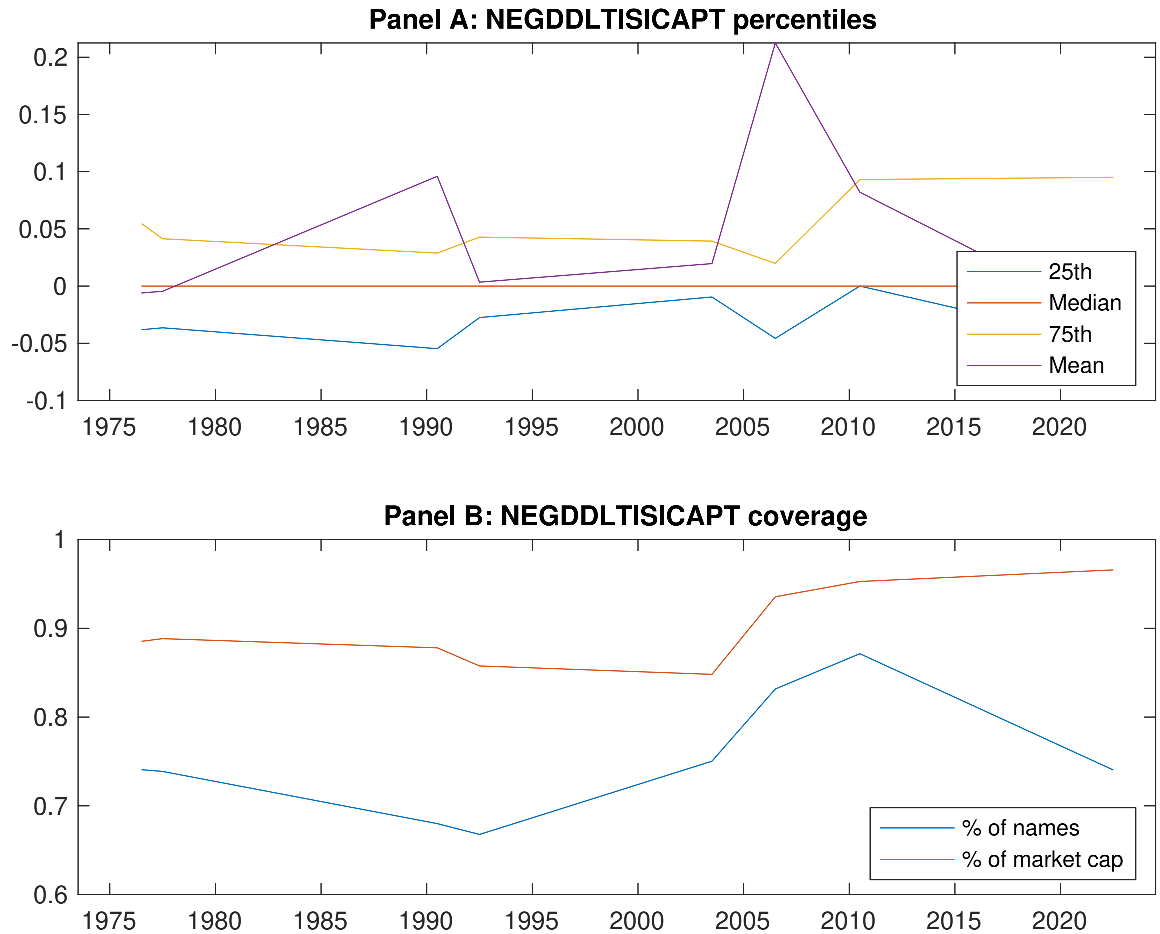


Figure 1: Times series of LEI percentiles and coverage.
This figure plots descriptive statistics for LEI. Panel A shows cross-sectional percentiles of LEI over the sample. Panel B plots the monthly coverage of LEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.66	0.64	0.72	0.83	0.27
	[2.62]	[3.68]	[3.24]	[4.11]	[4.13]	[3.73]
α_{CAPM}	-0.17	0.05	-0.03	0.13	0.15	0.32
	[-3.14]	[1.05]	[-0.43]	[2.68]	[2.70]	[4.33]
α_{FF3}	-0.17	0.01	0.03	0.10	0.15	0.32
	[-3.27]	[0.15]	[0.57]	[2.25]	[2.78]	[4.40]
α_{FF4}	-0.14	0.02	0.07	0.07	0.14	0.28
	[-2.65]	[0.58]	[1.17]	[1.56]	[2.51]	[3.77]
α_{FF5}	-0.17	-0.03	0.09	0.01	0.10	0.28
	[-3.29]	[-0.81]	[1.55]	[0.13]	[1.89]	[3.78]
α_{FF6}	-0.15	-0.02	0.11	-0.01	0.10	0.25
	[-2.85]	[-0.42]	[1.93]	[-0.23]	[1.79]	[3.39]
Panel B: Fama and French (2018) 6-factor model loadings for LEI-sorted portfolios						
β_{MKT}	1.08	0.98	0.98	0.96	1.03	-0.04
	[86.99]	[98.38]	[73.95]	[94.17]	[80.69]	[-2.49]
β_{SMB}	0.11	-0.12	-0.02	-0.03	0.13	0.02
	[5.94]	[-8.01]	[-1.07]	[-1.91]	[6.92]	[0.88]
β_{HML}	0.02	0.15	-0.11	-0.01	-0.10	-0.12
	[0.92]	[7.58]	[-4.31]	[-0.52]	[-4.12]	[-3.74]
β_{RMW}	0.10	0.10	-0.05	0.11	0.04	-0.06
	[4.13]	[5.30]	[-2.05]	[5.69]	[1.41]	[-1.93]
β_{CMA}	-0.12	0.02	-0.13	0.21	0.15	0.27
	[-3.33]	[0.65]	[-3.46]	[7.16]	[4.01]	[5.39]
β_{UMD}	-0.04	-0.03	-0.04	0.02	0.01	0.04
	[-2.94]	[-2.65]	[-2.68]	[2.45]	[0.56]	[2.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	526	1062	584	648	
me ($\$10^6$)	1494	3076	2226	3098	1458	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the LEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.32	0.32	0.28	0.28	0.25
			[3.73]	[4.33]	[4.40]	[3.77]	[3.78]	[3.39]
Quintile	NYSE	EW	0.18	0.21	0.20	0.18	0.19	0.17
			[3.99]	[4.58]	[4.29]	[3.78]	[3.97]	[3.63]
Quintile	Name	VW	0.26	0.32	0.33	0.28	0.27	0.24
			[3.71]	[4.57]	[4.74]	[3.97]	[3.92]	[3.44]
Quintile	Cap	VW	0.23	0.27	0.28	0.23	0.19	0.16
			[3.48]	[4.17]	[4.22]	[3.43]	[2.99]	[2.50]
Decile	NYSE	VW	0.31	0.36	0.35	0.36	0.23	0.25
			[3.38]	[3.92]	[3.79]	[3.81]	[2.47]	[2.70]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.22	0.25	0.25	0.23	0.22	0.20
			[2.92]	[3.42]	[3.46]	[3.13]	[2.97]	[2.72]
Quintile	NYSE	EW	-0.08					
			[-1.29]					
Quintile	Name	VW	0.21	0.26	0.27	0.25	0.23	0.20
			[2.86]	[3.76]	[3.89]	[3.50]	[3.26]	[2.97]
Quintile	Cap	VW	0.18	0.22	0.23	0.20	0.16	0.14
			[2.69]	[3.43]	[3.45]	[3.04]	[2.44]	[2.12]
Decile	NYSE	VW	0.24	0.29	0.28	0.29	0.18	0.18
			[2.57]	[3.09]	[2.98]	[3.02]	[1.92]	[1.97]

Table 3: Conditional sort on size and LEI

This table presents results for conditional double sorts on size and LEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LEI and short stocks with low LEI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		LEI Quintiles					LEI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.67	0.91	0.89	0.88	0.80	0.13	0.15	0.14	0.09	0.11	0.08
		[2.46]	[3.30]	[3.30]	[3.19]	[2.87]	[1.48]	[1.72]	[1.60]	[1.03]	[1.27]	[0.91]
	(2)	0.71	0.90	0.84	0.87	0.87	0.16	0.20	0.17	0.17	0.13	0.14
		[2.66]	[3.62]	[3.31]	[3.53]	[3.44]	[2.00]	[2.47]	[2.20]	[2.14]	[1.70]	[1.72]
	(3)	0.80	0.79	0.82	0.82	0.88	0.09	0.14	0.13	0.09	0.12	0.08
		[3.17]	[3.60]	[3.42]	[3.66]	[3.75]	[1.09]	[1.85]	[1.68]	[1.09]	[1.45]	[1.04]
	(4)	0.72	0.81	0.82	0.73	0.92	0.20	0.24	0.23	0.19	0.21	0.18
		[3.11]	[3.92]	[3.67]	[3.52]	[4.24]	[2.63]	[3.22]	[3.07]	[2.46]	[2.66]	[2.23]
	(5)	0.47	0.63	0.59	0.60	0.83	0.36	0.38	0.39	0.33	0.32	0.28
		[2.38]	[3.50]	[3.06]	[3.31]	[4.33]	[4.01]	[4.24]	[4.36]	[3.66]	[3.52]	[3.09]

Panel B: Portfolio average number of firms and market capitalization

		LEI Quintiles Average n					LEI Quintiles Average market capitalization ($$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	392	393	393	393	388	37	34	32	33	35
	(2)	107	107	107	107	107	61	62	60	61	61
	(3)	76	77	76	77	76	107	110	105	106	108
	(4)	64	65	64	65	64	229	241	229	234	231
	(5)	59	59	59	59	59	1452	2217	1819	2183	1506

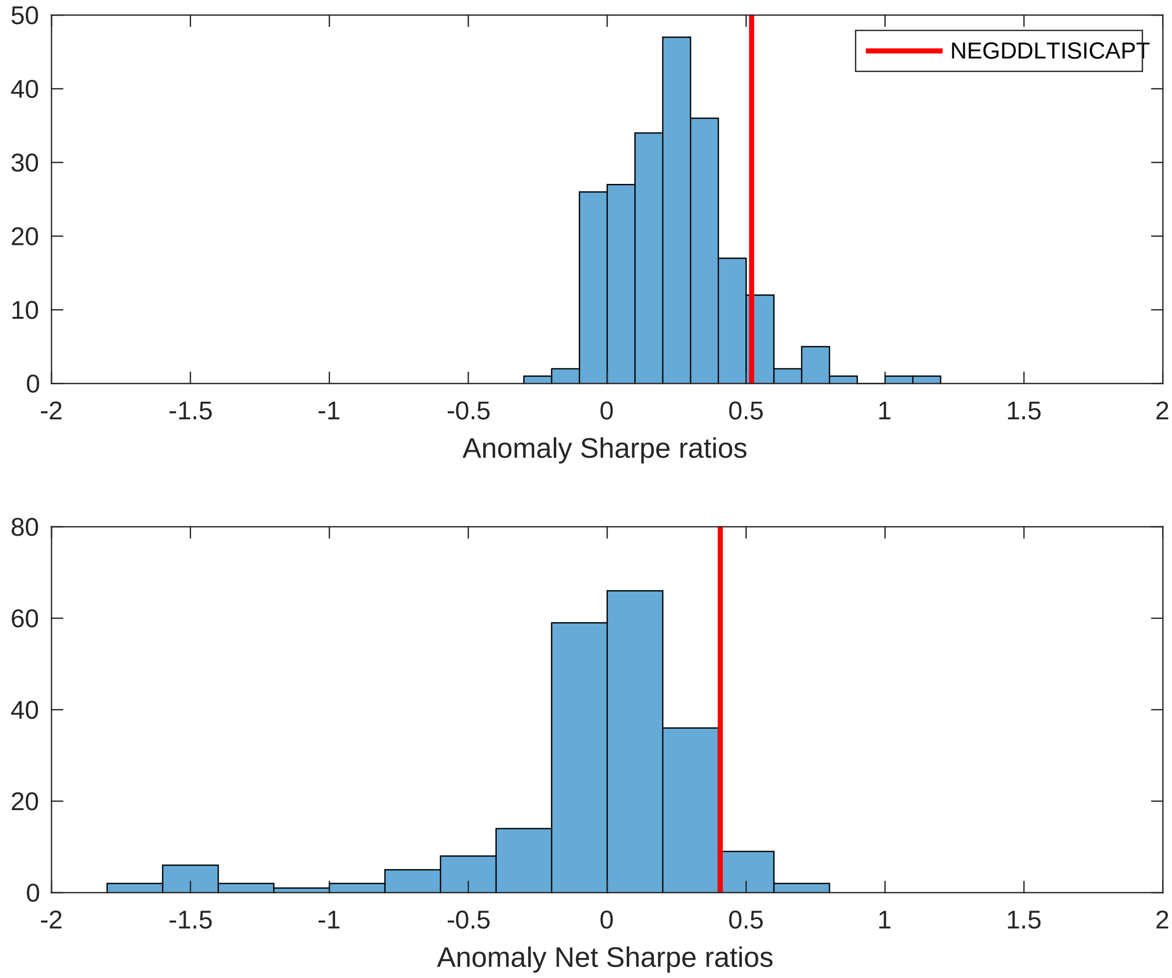


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

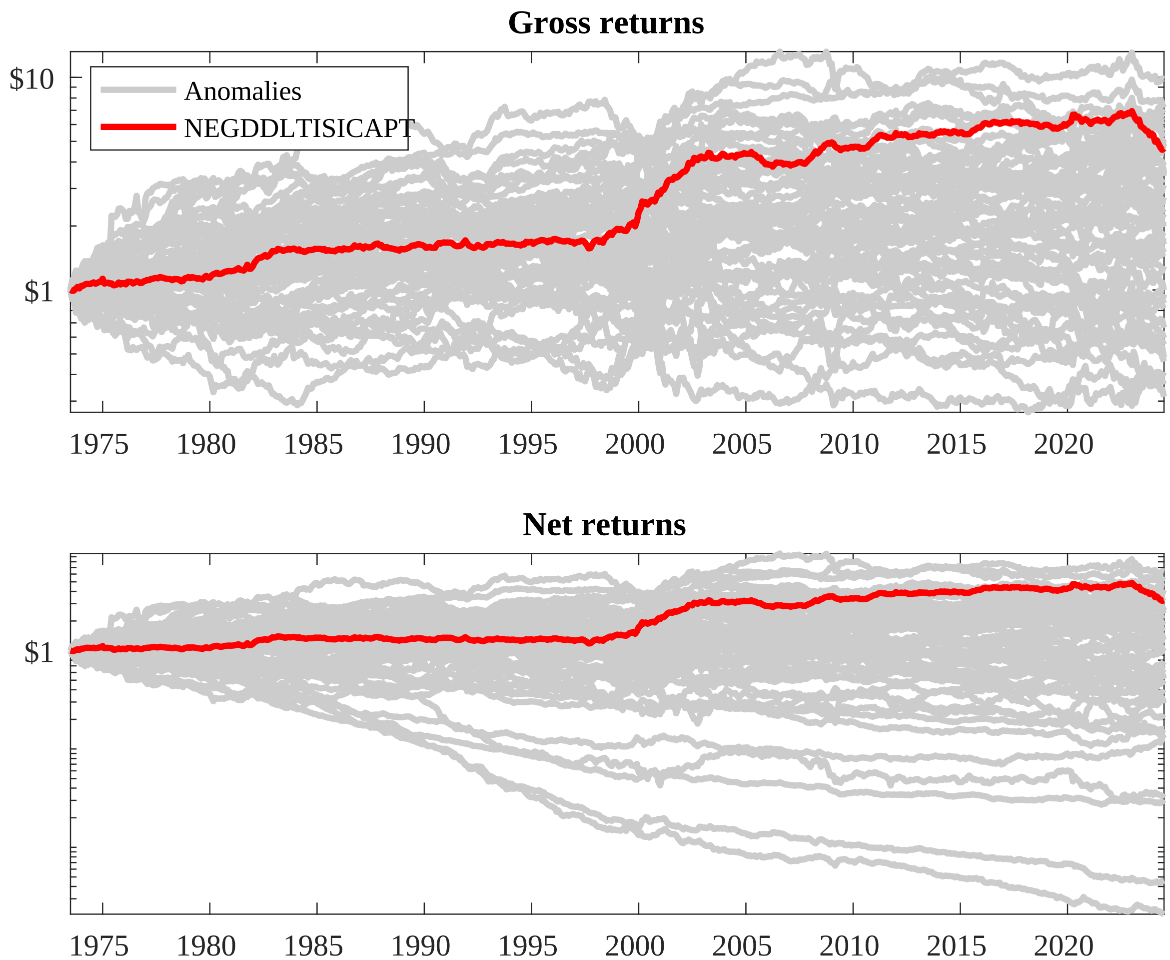
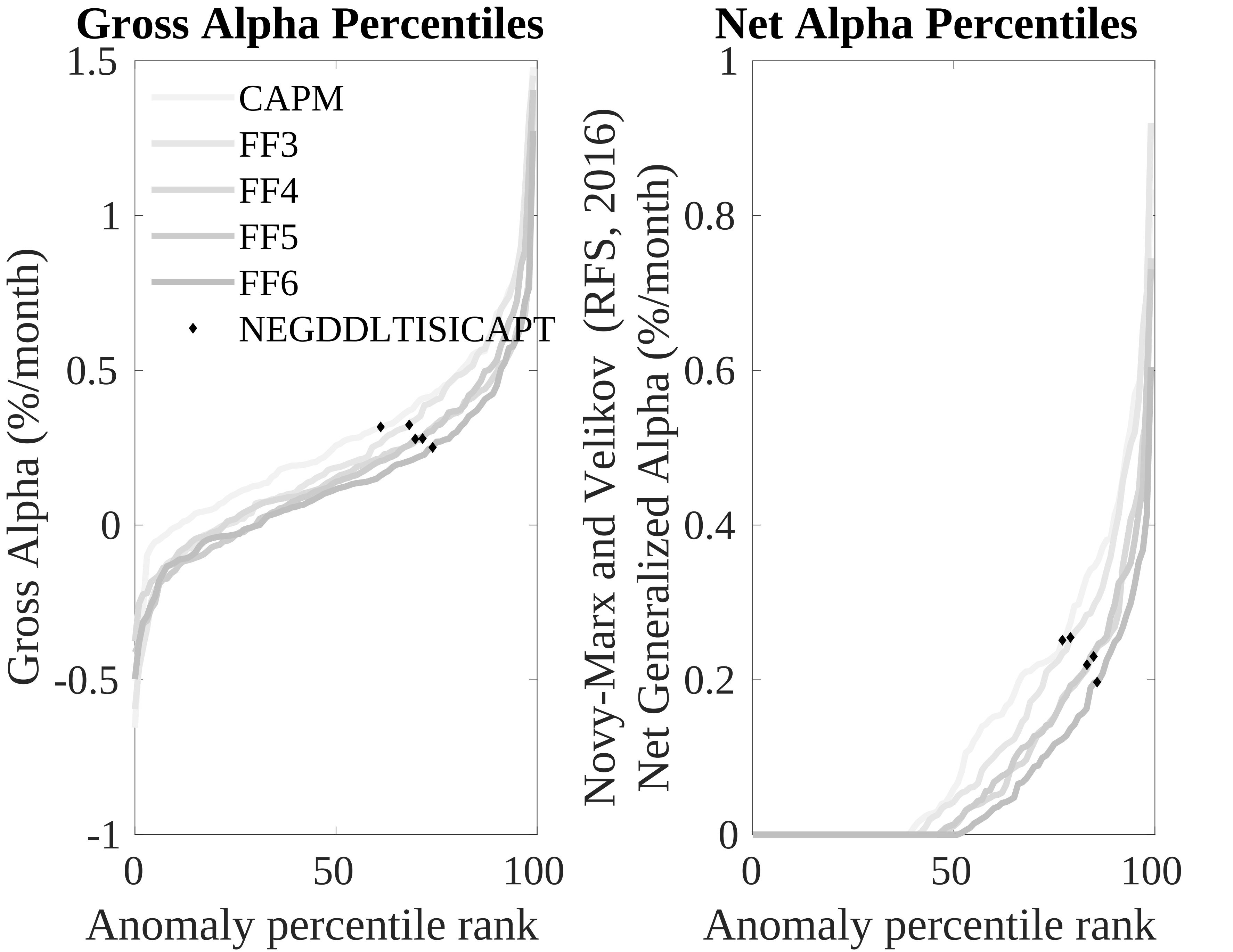


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISICAPT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

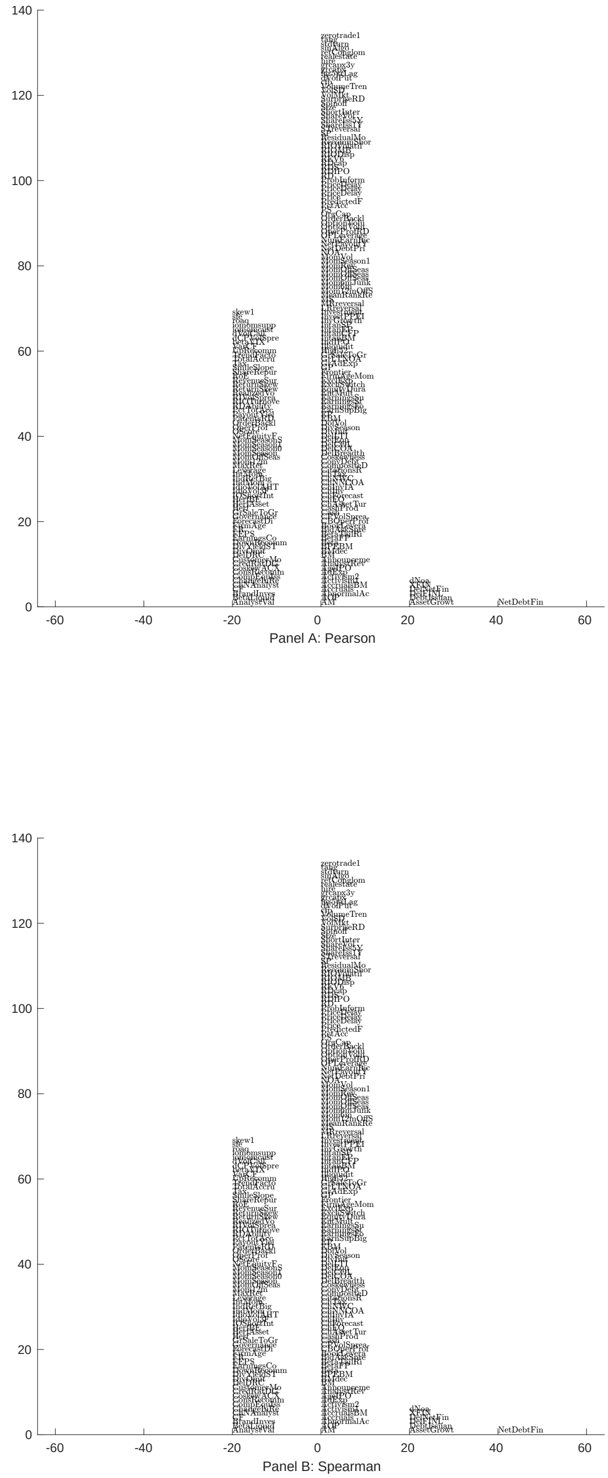


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with LEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

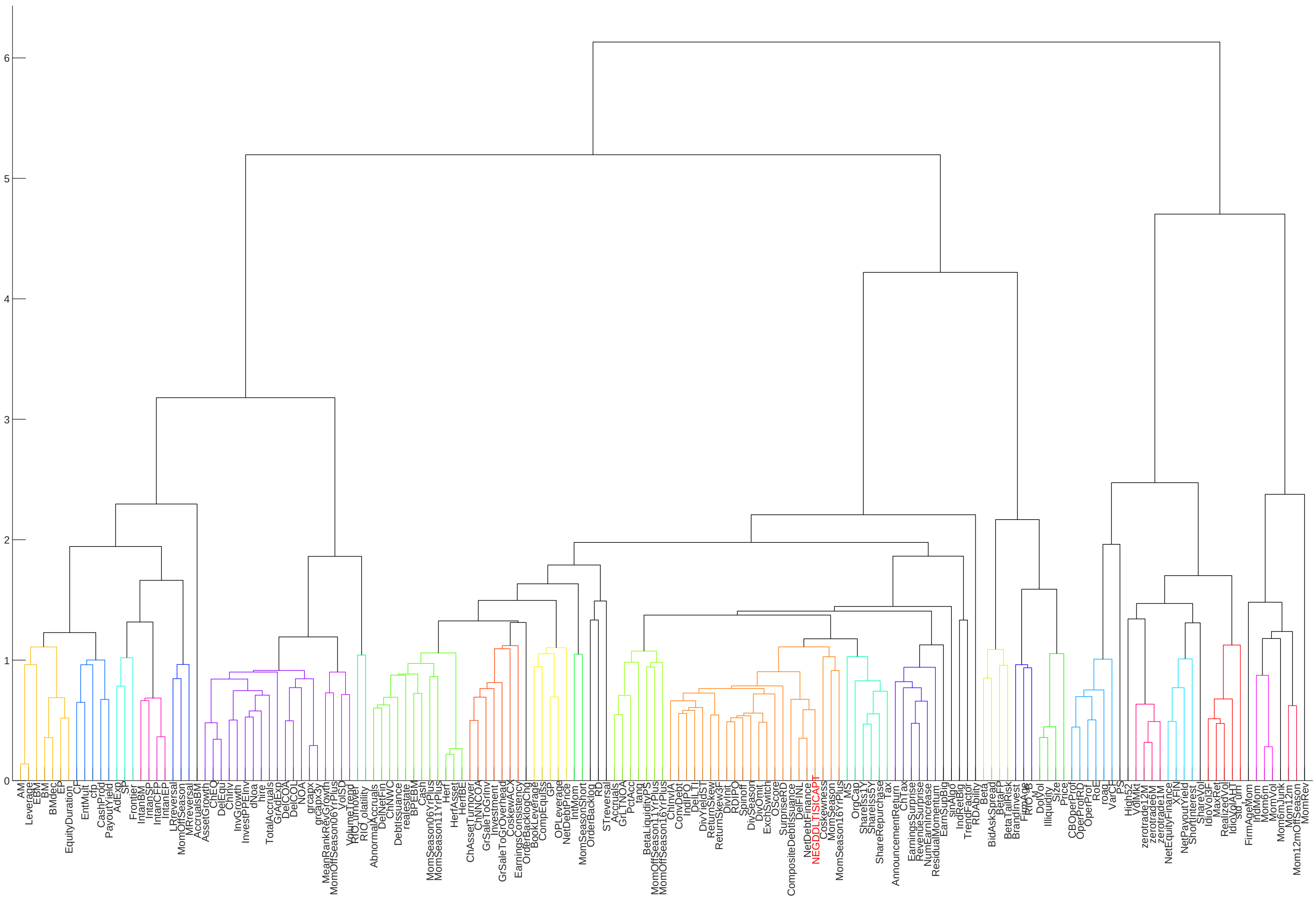


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

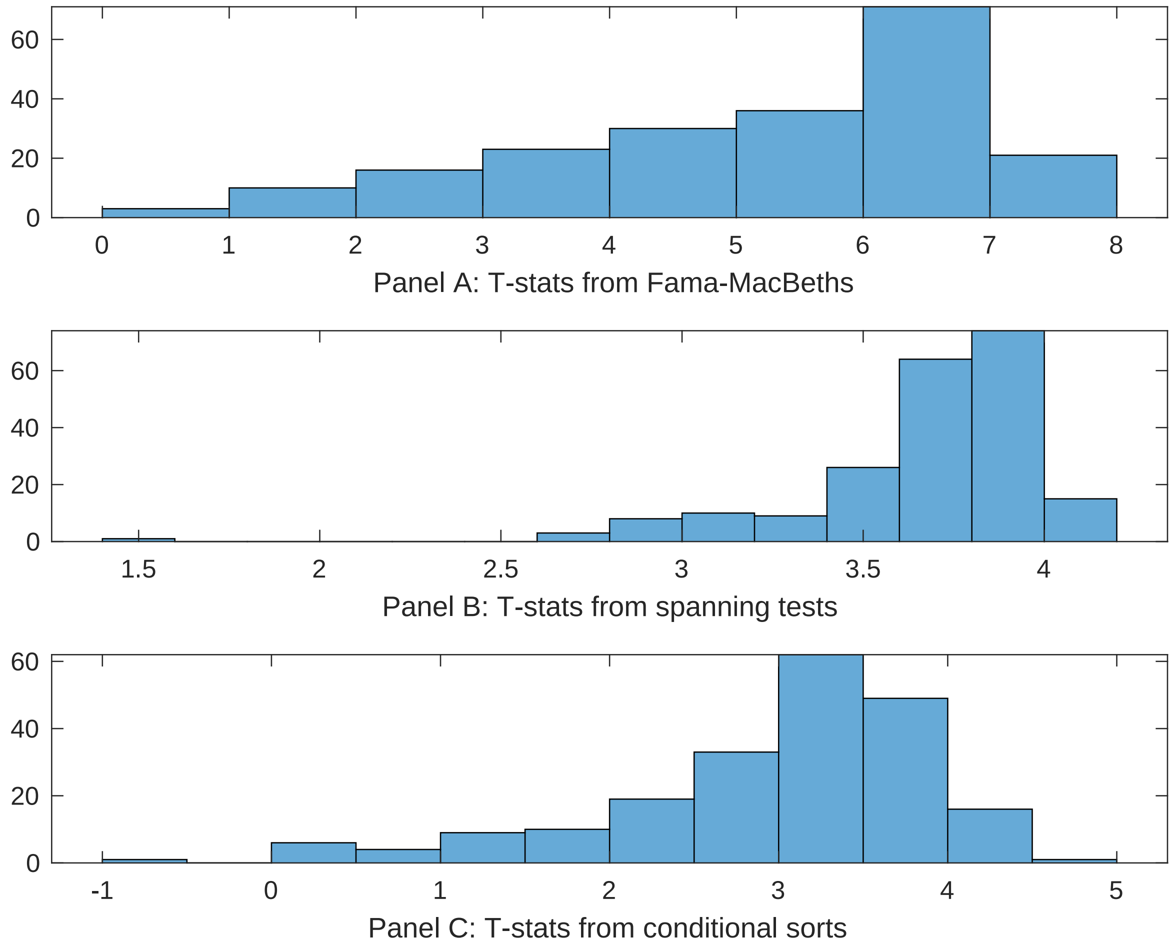


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of LEI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LEI} LEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LEI. Stocks are finally grouped into five LEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LEI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LEI} LEI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.13	0.15
	[5.37]	[5.33]	[5.32]	[5.74]	[5.69]	[5.25]	[5.72]
LEI	0.89	0.10	0.34	0.16	0.16	0.29	0.11
	[1.50]	[1.58]	[5.10]	[2.54]	[2.35]	[4.66]	[0.14]
Anomaly 1	0.17						-0.24
	[9.07]						[-0.05]
Anomaly 2		0.20					0.86
		[8.42]					[1.29]
Anomaly 3			0.40				0.69
			[6.86]				[1.19]
Anomaly 4				0.17			0.13
				[7.62]			[4.10]
Anomaly 5					0.19		0.13
					[6.09]		[2.36]
Anomaly 6						0.61	-0.52
						[4.03]	[-1.50]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the LEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LEI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.22	0.22	0.24	0.24	0.22	0.20	0.19
	[3.00]	[3.07]	[3.32]	[3.26]	[3.02]	[2.78]	[2.67]
Anomaly 1	18.41						2.01
	[4.47]						[0.35]
Anomaly 2		21.09					10.96
		[5.34]					[2.05]
Anomaly 3			8.03				7.25
			[2.81]				[2.48]
Anomaly 4				7.03			-1.94
				[2.12]			[-0.55]
Anomaly 5					14.86		11.33
					[4.07]		[2.98]
Anomaly 6						17.72	13.32
						[5.17]	[3.44]
mkt	-3.54	-3.92	-4.11	-4.06	-1.92	-3.63	-2.17
	[-2.11]	[-2.36]	[-2.43]	[-2.40]	[-1.10]	[-2.18]	[-1.26]
smb	0.61	0.63	3.38	2.52	7.27	3.53	6.51
	[0.24]	[0.25]	[1.31]	[0.99]	[2.60]	[1.40]	[2.26]
hml	-10.91	-11.28	-12.39	-12.88	-10.48	-12.45	-10.61
	[-3.37]	[-3.52]	[-3.80]	[-3.92]	[-3.22]	[-3.88]	[-3.30]
rmw	-7.89	-8.55	-5.07	-6.11	-15.12	-4.11	-11.78
	[-2.40]	[-2.61]	[-1.52]	[-1.84]	[-3.83]	[-1.25]	[-2.98]
cma	21.24	21.94	20.26	21.83	17.71	31.93	15.06
	[4.18]	[4.41]	[3.65]	[3.89]	[3.25]	[6.45]	[2.37]
umd	2.91	3.14	3.68	4.49	4.40	4.20	2.56
	[1.71]	[1.88]	[2.16]	[2.67]	[2.64]	[2.54]	[1.52]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	14	11	10	12	13	17

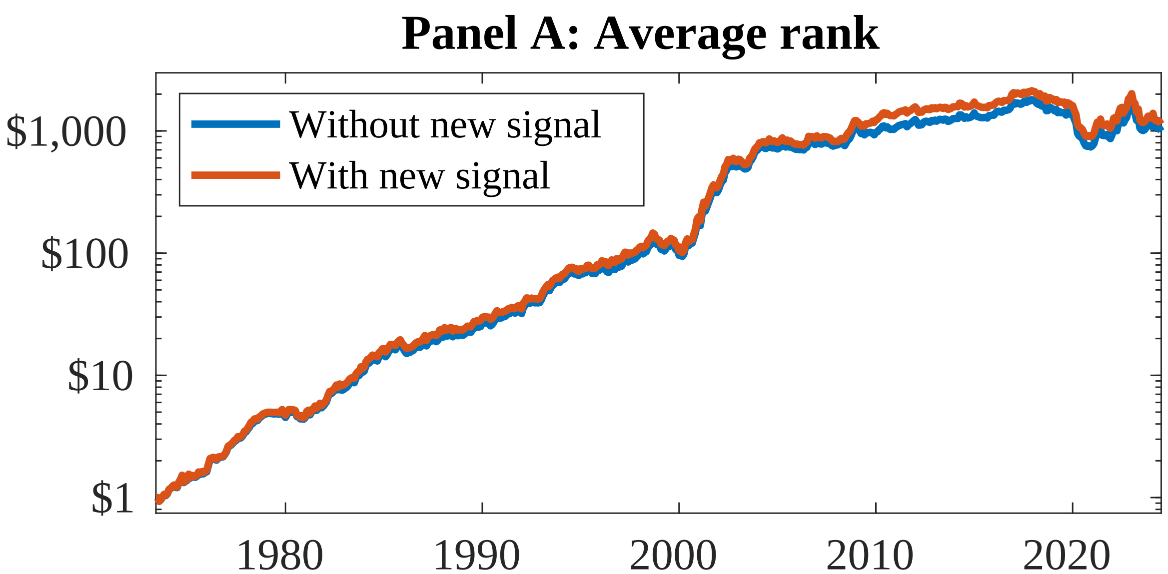


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LEI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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